

HYDROGENATED NITRILE BUTADIENE RUBBER (HNBR): HYDROGENATION PROCESSES, CATALYST SYSTEMS, AND POST-PROCESSING TRENDS ACROSS PRIMARY PATENTS

1. ABSTRACT

This report synthesizes four primary patents focused on hydrogenated nitrile butadiene rubber (HNBR) and closely related manufacturing routes. The target material is HNBR itself, including liquid HNBR, low-molecular-weight HNBR, and HNBR compositions, with emphasis on how hydrogenation is performed, what catalyst systems are used, and how residual catalyst or polymer properties are managed. Across the set, the dominant transformation is catalytic hydrogenation of unsaturated nitrile rubber (NBR) in solution, with one patent also disclosing simultaneous hydrogenation and metathesis to reduce molecular weight. Another patent focuses on producing HNBR with improved ageing properties by controlling noble-metal catalyst type and loading, while the most recent patent addresses liquid HNBR and appears to rely on hydrogenation chemistry and catalyst-related processing rather than worked examples.

The landscape is centered on noble-metal catalysis, especially ruthenium, palladium, and rhodium compounds, with explicit attention to catalyst residue management because residual metals can impair ageing, discoloration, and product performance. Solution polymerization/hydrogenation in organic solvent is the core medium in the disclosed preparation routes; water is not part of the hydrogenation medium in the extracted evidence, though downstream washing/coagulation is mentioned in prior-art discussion within one patent. The patents collectively show a strong trend toward process simplification: avoiding elaborate catalyst-removal steps, tuning catalyst loading to balance hydrogenation rate and ageing stability, and in one case using catalysts that enable concurrent hydrogenation and metathesis to obtain lower molecular weight HNBR with narrower molecular-weight distribution.

Key disclosed process windows include hydrogenation temperatures of 60–200°C, pressures of 700,000–15,000,000 Pa, and reaction times of 1–24 hours in one patent; another uses monochlorobenzene solution at 130–138°C and 85 bar with 100 psi H₂ degassing. Example data show that catalyst identity and loading strongly affect residual double bond reduction, molecular weight, and Mooney viscosity. Overall, the evidence supports solution-phase catalytic hydrogenation of NBR precursors as the principal route to HNBR, with catalyst selection and

post-reaction handling being the main levers for product quality.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP4215556A4
- Patent Title: LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF
- Assignee: Zannan Scitech Co., Ltd.
- Jurisdiction: EP
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a process for making liquid hydrogenated nitrile butadiene rubber and its uses, with the detailed description indicating hydrogenation-related treatment of HNBR and catalyst-focused chemistry. The extracted evidence does not include worked examples or a fully specified reaction recipe, so the synthesis is only identifiable at a high level as HNBR hydrogenation/process development. - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Not clearly disclosed in the extracted evidence

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - No worked-example sections were present in the patent text; report evidence falls back to claims/description. - Detailed Description: EP4215556A4 - LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF - Google Patents LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF Download PDF Info Publicatio

Technical Relevance This patent is directly relevant because it targets liquid HNBR, a specific HNBR embodiment, and is classified under hydrogenation of rubber. Even though the extracted evidence is sparse, it is clearly within the HNBR manufacturing space.

Patent 2

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses hydrogenation of at least partially unsaturated nitrile rubber (NBR) in solution using ruthenium, palladium, or rhodium compounds, with the stated goal of producing HNBR having good ageing properties. The process emphasizes catalyst residue control rather than a novel polymerization route, and the claims define temperature, pressure, time, and catalyst loading windows for the hydrogenation step. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700 000 pascals to 15 000 000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - HNBR composition: 36% by weight acrylonitrile units — Example materials (Therban 3627) - Mooney viscosity: 66 ML 1+4@100°C — Example materials (Therban 3627) - Gel content: 0.7% — Example materials (Therban 3627) - Noble metal content: 2 ppm rhodium, <1 ppm ruthenium, <1 ppm palladium — Example materials (Therban 3627) - Ageing temperature: 140°C — Example ageing procedure - Ageing time: 4 days — Example ageing procedure - Acetone concentration: 10% — Example procedure - Mixing time: 2 hours — Example procedure - Drying temperature: 55°C — Example procedure - Medium & Water Role: The core hydrogenation medium is solution-phase NBR in solvent; no water-related process step is disclosed in the extracted evidence for the claimed process. Water appears only in prior-art discussion as part of cited washing/coagulation methods, not as the disclosed invention process. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - The core hydrogenation medium is solution-phase NBR in solvent; no water-related process step is disclosed in the extracted evidence for the claimed process. Water appears only in prior-art discussion as part of cited washing/coagulation methods, not as the disclosed invention process. - Core reaction medium: Solution polymerization/hydrogenation in organic solvent

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Ageing tests were performed on HNBR compositions containing different noble-metal compounds, showing the effect on gel content and

Mooney viscosity after ageing. - Example 2: Ruthenium compound at 42 ppm gave only modest ageing-related changes relative to the reference. - Example 7: Palladium acetate at 44 ppm was associated with larger ageing changes than the ruthenium cases. - Example 13: Rhodium acetate at 79 ppm was tested as part of the comparative ageing study. - Example 17: Rhodium acetate at 280 ppm was included to probe higher noble-metal loading effects.

Technical Relevance This patent is highly relevant because it directly claims a solution hydrogenation route from NBR to HNBR and specifies catalyst classes and operating windows. It also links catalyst choice and residue level to ageing performance, which is a key quality issue in HNBR manufacture.

Patent 3

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Arlanxeo Deutschland Gmbh
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a method for evaluating the effect of ruthenium, palladium, and rhodium compounds on the ageing behavior of HNBR compositions, rather than the initial hydrogenation synthesis itself. The examples dissolve commercial HNBR in acetone, add noble-metal compounds, dry the compositions, and then age them to compare gel content and Mooney viscosity changes. - HNBR acrylonitrile content: 36% by weight — Example materials (Therban 3627) - Mooney viscosity: 66 ML 1+4@100°C — Example materials (Therban 3627) - Gel content: 0.7% — Example materials (Therban 3627) - Noble metal content: 2 ppm rhodium, <1 ppm ruthenium, <1 ppm palladium — Example materials (Therban 3627) - Acetone concentration: 10% — Example procedure - Mixing time: 2 hours — Example procedure - Drying temperature: 55°C — Example procedure - Ageing temperature: 140°C — Example ageing procedure - Ageing time: 4 days — Example ageing procedure - Gel content measurement solvent: approximately 20 ml methyl ethyl ketone — Gel Content section - Drying temperature for gel measurement: 60°C — Gel Content section - Centrifugation speed: 25 000 rpm — Gel Content section - Medium & Water Role: No water-related process step disclosed in the extracted evidence. The disclosed examples use acetone solution, drying, and ageing of HNBR compositions, with no aqueous polymerization or water workup described. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. The disclosed examples use acetone solution, drying, and ageing of HNBR compositions, with no aqueous polymerization or water workup described. - Core reaction medium: Organic solvent solution for HNBR composition preparation; not a hydrogenation medium

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1: Commercial HNBR was used as the base polymer for ageing studies with noble-metal additives. - Example 1: The HNBR was dissolved in acetone, mixed with catalyst compounds, dried, and aged for 4 days at 140°C. - Example 1: Gel content and Mooney viscosity were measured before and after ageing to assess catalyst effects.

Technical Relevance This patent is relevant to HNBR because it uses HNBR as the target material and quantifies how noble-metal residues affect ageing stability. However, it does not disclose the hydrogenation synthesis of HNBR itself, so its relevance is indirect for synthesis and direct for post-synthesis property retention.

Patent 4

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: LANXESS Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a process for producing low molecular weight HNBR by simultaneous hydrogenation and metathesis of nitrile rubber in the presence of ruthenium or osmium compounds. The process is carried out in organic solvent under hydrogen, optionally with a co-olefin, and is designed to reduce molecular weight while hydrogenating the polymer backbone. - Acrylonitrile content: 34% by weight — Example 1 starting NBR - Mooney viscosity: 28 Mooney units — Example 1 starting NBR - Residual moisture content: <0.5% by weight — Example 1 starting NBR - Mw: 214 000 g/mol — Example 1 starting NBR - Mn: 67 000 g/mol — Example 1 starting NBR - PDI: 3.2 — Example 1 starting NBR - NBR amount: 518 grams — Example 1 - Monochlorobenzene amount: 4300 grams — Example 1 - Solution concentration: 12% — Example 1 - Degassing cycles with H₂: 3 times — Example 1 - Hydrogen degassing pressure: 100 psi — Example 1 - Reactor agitation: 600 rpm — Example 1 - Temperature before catalyst addition: 130°C — Example 1 - Pressure: 85 bar — Example 1 - Reaction temperature: 138°C — Example 1 - Catalyst amount: 0.041 phr — Example 1, Hoveyda catalyst - Catalyst amount: 0.045 phr — Example 1, Arlt catalyst - Catalyst amount: 0.060 phr — Example 1, Wilkinson's catalyst - Metal content: 65.6 ppm — Example 1 - Co-catalyst amount: 1.0 phr — Example 1, Wilkinson's catalyst - Reaction time: 0 to 240 min — Example 1, hydrogenation profile - Residual double bond: 100% at 0 min — Example 1 - Residual double bond: 2.5% at 60 min — Example 1, HNBR 1 - Residual double bond: 0.6% at 120 min — Example 1, HNBR 1 - Residual double bond: 0.4% at 240 min — Example 1, HNBR 1 - Mw: 113371 g/mol — Example 1, HNBR 1 after 240 min - Mn: 48352 g/mol — Example 1, HNBR 1 after 240 min - PDI: 2.34 — Example 1, HNBR 1 after 240 min - Mooney viscosity: 33.3 — Example 1, HNBR 1 after 240 min -

Mw: 133758 g/mol — Example 1, HNBR 2 after 240 min - Mn: 54595 g/mol — Example 1, HNBR 2 after 240 min - PDI: 2.45 — Example 1, HNBR 2 after 240 min - Mooney viscosity: 41.3 — Example 1, HNBR 2 after 240 min - Mw: 195304 g/mol — Example 1, HNBR 3 after 240 min - Mn: 64130 g/mol — Example 1, HNBR 3 after 240 min - PDI: 3.05 — Example 1, HNBR 3 after 240 min - Mooney viscosity: 69.3 — Example 1, HNBR 3 after 240 min - Medium & Water Role: The core reaction medium is an organic solvent solution in monochlorobenzene under hydrogen; no water-related process step is disclosed in the extracted evidence. The only water-like term in the evidence is residual moisture content of the starting NBR, which is not a process water step. - degree of hydrogenation: Residual double bond: 0.4% at 240 min — Example 1, HNBR 1

Medium & Water Role - The core reaction medium is an organic solvent solution in monochlorobenzene under hydrogen; no water-related process step is disclosed in the extracted evidence. The only water-like term in the evidence is residual moisture content of the starting NBR, which is not a process water step. - Core reaction medium: Organic solvent solution polymerization/hydrogenation in monochlorobenzene

degree of hydrogenation - Residual double bond: 0.4% at 240 min — Example 1, HNBR 1

Relevant Experimental Evidence - Example 1: Hoveyda, Arlt, and Wilkinson catalysts were compared for simultaneous hydrogenation/metathesis of NBR in monochlorobenzene. - Example 1: HNBR 1 and HNBR 2 showed faster reduction of residual double bonds than the Wilkinson comparison catalyst. - Example 1: The inventive catalysts produced lower molecular weight HNBR with narrower PDI than the comparison catalyst. - Example 1: HNBR 1 reached 0.4% residual double bond after 240 min and had Mw 113371 g/mol. - Example 1: HNBR 1 and HNBR 2 had substantially lower Mooney viscosity than HNBR 3 after 240 min.

Technical Relevance This patent is directly relevant because it produces HNBR from NBR and explicitly combines hydrogenation with molecular-weight reduction. It provides concrete process conditions and shows how catalyst choice controls residual unsaturation and final polymer properties.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	degree of hydrogenation	Key Findings
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EP4215556A4	Zannan Scitech Co., Ltd.	2024	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	No worked-example sections were present in the patent text; report evidence falls back to claims/description.
EP3827029A1	ARLANXEO Deutschland GmbH	2021	The core hydrogenation medium is solution-phase NBR in solvent; no water-related process step is disclosed in the extracted evidence for the claimed process. Water appears only in prior-art discussion as part of cited washing/coagulation methods, not as the disclosed invention process.	Not disclosed in extracted evidence	Example 1: Ageing tests were performed on HNBR compositions containing different noble-metal compounds, showing the effect

US20210340285A1	Arlanxo Deutschland GmbH	2021	No water-related process step disclosed in the extracted evidence. The disclosed examples use acetone solution, drying, and ageing of HNBR compositions, with no aqueous polymerization or water workup described.	Not disclosed in extracted evidence	Example 1: Commercial HNBR was used as the base polymer for ageing studies with noble-metal additives.
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EP2473281B1	LANXESS Deutschland GmbH	2014	The core reaction medium is an organic solvent solution in monochlorobenzene under hydrogen; no water-related process step is disclosed in the extracted evidence. The only water-like term in the evidence is residual moisture content of the starting NBR, which is not a process water step.	Residual double bond: 0.4% at 240 min — Example 1, HNBR 1	Example 1: Hoveyda, Arlt, and Wilkinson catalysts were compared for simultaneous hydrogenation/metathesis of NBR in mono
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SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: LANXESS Deutschland GmbH
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This patent is about nitrile rubber production and optionally hydrogenation, but the central material is nitrile rubber rather than HNBR as the requested base material.

Supporting Patent 2

- Patent Number: US8530584B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Lanxess Deutschland Gmbh
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about catalyst systems for metathesis of nitrile rubber, which is related precursor chemistry but not HNBR as the primary invention.

Supporting Patent 3

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Lanxess Deutschland Gmbh
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about catalyst systems for metathesis of nitrile rubber, which is related to HNBR precursor chemistry but not about HNBR itself as the central invention.

Supporting Patent 4

- Patent Number: EP3294574B1
- Patent Title: Improvement of the roll resistance of diene rubber tires by means of silane- ...
- Assignee: Evonik Operations GmbH
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about silane-modified polybutadienes in rubber mixtures, not HNBR as the central material.

Supporting Patent 5

- Patent Number: EP1994090B1
- Patent Title: Process for producing compatibilized polymer blends and articles
- Assignee: PRS Mediterranean Ltd.
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: HNBR is only one of many possible polymers in a compatibilized blend process, so the invention is about blend compatibilization rather than HNBR itself.

Supporting Patent 6

- Patent Number: US8557921B2

- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Dow Global Technologies Llc
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: Although partially hydrogenated butadiene nitrile rubber is one of the polymer blocks, the invention is about metathesis-derived meta-block copolymers, not HNBR as the primary material.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- All four patents are centered on HNBR, but only EP2473281B1 explicitly discloses a quantitative hydrogenation outcome via residual double bond (0.4% for HNBR 1 after 240 min).
- EP3827029A1 and US20210340285A1 focus on noble-metal residue effects and ageing behavior of HNBR rather than the hydrogenation extent itself; both use solution-based organic media and emphasize catalyst identity/loading.
- EP2473281B1 uses monochlorobenzene solution at 85 bar and 138°C with simultaneous hydrogenation/metathesis, whereas EP3827029A1 claims broader hydrogenation windows of 60–200°C, 700,000–15,000,000 Pa, and 1–24 h.
- Catalyst systems differ: EP3827029A1 centers on Ru/Pd/Rh compounds with ppm-level loadings; EP2473281B1 compares Hoveyda, Arlt, and Wilkinson-type ruthenium catalysts with phr-level loadings; US20210340285A1 studies Ru/Pd/Rh additives in finished HNBR compositions.
- Water is not part of the core reaction medium in any of the extracted inventive examples; where water-related operations appear, they are only in prior-art discussion or absent entirely.
- EP2473281B1 is the only patent in the set that clearly links catalyst choice to both hydrogenation kinetics and molecular-weight reduction, making it the most process-specific HNBR synthesis disclosure in the group.

4. CONCLUSION

The extracted evidence shows that HNBR is predominantly made by solution-phase catalytic hydrogenation of NBR in organic solvent, with noble-metal catalysts being the dominant technology family. The strongest synthesis disclosure is EP2473281B1, which demonstrates simultaneous hydrogenation and metathesis in monochlorobenzene under hydrogen, yielding low-molecular-weight HNBR with low residual double bond content. EP3827029A1 provides a broader industrial hydrogenation window and highlights the importance of controlling Ru/Pd/Rh catalyst residues to preserve ageing performance, while US20210340285A1 shows that residual noble metals in HNBR compositions measurably affect ageing behavior. EP4215556A4 is relevant to liquid HNBR but the extracted evidence is too sparse to recover detailed process parameters.

Based on the evidence, the most defensible synthesis recommendation is solution hydrogenation of NBR in an organic solvent with carefully selected noble-metal catalyst systems, controlled temperature and pressure, and explicit attention to catalyst residue management to balance hydrogenation rate, residual unsaturation, molecular weight, and ageing stability.

5. REFERENCES

- EP4215556A4 | LIQUID HYDROGENATED NITRILE BUTADIENE RUBBER, PROCESS THEREOF AND USE THEREOF | Zannan Scitech Co., Ltd. | EP | 2024 | <https://patents.google.com/patent/EP4215556A4/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | ARLANXEO Deutschland GmbH | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Arlanxeo Deutschland GmbH | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | LANXESS Deutschland GmbH | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>